Q.1. Explain how energy flow supports the second law of thermodynamics in an ecosystem.

A. As per the second law of thermodynamics, every activity that includes transformation includes wastage of energy in the form of heat and rise in disorganization excluding deep hydrothermal ecosystems. Out of the total PAR (Photosynthetically Active Radiation), only 2-10% is absorbed by organisms involved in photosynthesis to produce organic matter. Furthermore, the energy is utilized in metabolic activities, to form food and in storing biomass (very less). Biomass or the trapped energy is passed to the next trophic level as per the Lindeman's law. 10% of the stored energy is transferred from one to the next consecutive trophic level.

Q.2. Answer the following questions. Write the outcomes of the following events.

a) The consequence of eliminating all producers

b) The consequence of eliminating all entities at the herbivore level

c) The consequence of eliminating all top carnivore entities

A.5. a) It diminishes primary production in an ecosystem and hence unavailability of biomass to higher trophic levels

b) It would result in an increase in primary productivity and biomass of producers. Carnivorous animals, due to unavailability of food, will not survive.

c) There will be an increase in the herbivore population, resulting in over-grazing and hence desertification.

Question 03. Rajni was watching a programme on TV about the earth. The host was explaining how the increased cutting of trees and forests is affecting the ecosystem, its components and services, etc. She got curious and asked her teacher about 'what services the ecosystem provides to human beings?

(i) How does ecosystem services benefit us?

(ii) Name the value put forth by Robert Constanza and his colleagues on ecosystem services.

(iii) What values are showed by Rajni?

Answer: (i) Healthy ecosystems provide wide range of economic, environmental, aesthetic goods and services. These services include purification of air, mitigation of droughts and floods, cycling of nutrients, providing habitats for wildlife, maintenance of biodiversity, etc.

(ii) Robert Constanza and his colleagues have estimated the average value of fundamental ecosystem service at US $ 33 trillion a year.

(iii) Values shown by Rajni are awareness and curiosity towards environmental affairs.

Questions 04: (i) Explain the significance of ecological pyramids with the help of an example.

(ii) Why are the pyramids referred to as upright or inverted? Explain.

Answer: (i) Significance of ecological pyramids: They graphically represent the relation between producers and consumers in order to calculate energy content, biomass and number of organisms of that trophic level. A trophic level represents only a functional level not a species as such. A given species may occupy more than one trophic level in the same ecosystem at the same time. The ecological pyramids provide an overall idea of the trophic levels occupied by an organism in an ecosystem. Example A sparrow is a primary consumer when it eats seeds, fruits, peas and a secondary consumer when it eats insects and worms.

(ii) Upright pyramids: When producers are more in number and biomass than the herbivores and herbivores are more in number and biomass than the carnivores, pyramids are referred to as upright. Energy at a lower trophic level is always more than at a higher trophic level. Pyramid of energy is referred to as always upright.

Inverted pyramids When the numbers and biomass of producers are less and consumers increase and become largest in top consumer level, Pyramid of number and biomass is referred to as inverted.

Question 05. (i) With suitable examples, explain the energy flow through different trophic levels. What does each bar in this pyramid represent?
(ii) Write any two limitations of ecological pyramids.
Answer: (i) The energy flows unidirectionally from the first trophic level (producers) to last trophic level (consumers) and as the energy flows from one trophic level to another, some energy is always lost as heat into the surrounding environment. So, the amount of energy flowing decreases at each successive trophic level. This can be explained with the help of a diagram of a grazing food chain. The pyramid of energy is always upright and each bar in the pyramid indicates the amount of energy present at each trophic level in a given time or per unit area.
(ii) The limitations of ecological pyramids are:
It does not consider the same single species operating at two or more trophic levels.
It assumes simple food chains that do not exist in nature and do not accommodate food web.
Saprophytes, detritivores and decomposers are not given any place in pyramids, despite their vital role in ecosystem (any two).

Que 06. Why is biodiversity important to the planet?
Ans. Biodiversity is important to the planet, because healthy, diverse ecosystems are better equipped to withstand and recover from a variety of disasters. All the species in an area work together to survive and maintain their ecosystem. For instance, large carnivores, such as lions, leopards, and wolves are essential for a healthy ecosystem. If the population of such carnivores falls, the herbivores they prey upon will likely see a population increase. This results in environmental deterioration, because they graze more. Thus a healthy ecosystem depends on a delicate balance based on cooperation and mutual survival.

Questions 07: Explain in-situ and ex-situ conservation of biodiversity?
Answer: The in-situ method of conservation is done in the natural ecosystem or habitat. Examples of In-situ include national parks and wildlife sanctuaries.
The Ex-situ method of conservation is carried out on man-made habitats or ecosystems. Examples of Ex-situ include zoological gardens, seed banks, and gene banks.

See in notes also